

BLE Communication Protocol — Smart Pill Bottle

1. Advertisement Name

The device broadcasts as "Medical Vial App".

2. Service & Characteristic UUIDs

	Service A	Service B
Service UUID (16-bit)	0x00FF	0x00EE
Characteristic UUID	0xFF01	0xEE01
Descriptor UUID	0x3333	0x2222

128-bit UUID bases (for scanning/filtering):

- Service A: 5f9b349b-0080-8000-0010-0000FF000000
- Service B: 5f9b349b-0080-8000-0010-0000EE000000

3. Request Sync Command

The sync command opcode is 0x30 (not 0xB4 — that's SET_PILL_INTERVAL).

Full sync flow:

1. App writes [0x30][32-byte Auth Key] to trigger log sync
2. Device sends the marker string "SYNC_DATA" followed by a JSON payload
3. App writes [0x31][32-byte Auth Key] to ACK — device then deletes the log file

JSON log format received:

```
[
  {"ts": 1234567890, "type": "Normal", "evnt": "PILL_CHANGE", "chng": 1, "cnt": 19},
  {"ts": 1234567891, "type": "Tamper", "evnt": "OUTSIDE_WINDOW_CHANGE"}
]
```

4. Auth Key Format

- 32 bytes (256-bit) raw binary, sent immediately after the 1-byte opcode
- General payload structure: [Opcode (1 byte)][Auth Key (32 bytes)][Optional params...]

Examples:

Command	Payload
Lock	[0x10][32-byte key]
Set time	[0x32][32-byte key][4-byte unix timestamp LE]
WiFi creds	[0x14][32-byte key][\"SSID\", \"PASSWORD\"]
Pill interval	[0xB4][32-byte key][1-byte hours]

The auth key is first established via BIND_DEVICE (0x12).

5. Full Command Reference

Opcode	Name	Description
0x10	LOCK_CMD	Lock/unlock device
0x11	SET_THRESHOLD	Set refill threshold
0x12	BIND_DEVICE	Pair with auth key
0x13	UNBIND_DEVICE	Unpair device
0x14	SET_WIFI_CRED	Set WiFi SSID & password
0x15	SET_COMM_PROT	Set communication protocol
0x16	SET_API_TARGET	Set API/MQTT target
0x20	START_CALIB	Start load cell calibration
0x21	CALIB_PILL_ADD	Add pills during calibration
0x22	FINISH_CALIB	End calibration
0x30	REQUEST_SYNC	Trigger log sync
0x31	ACK_SYNC	Acknowledge sync done
0x32	SET_TIME	Set RTC time
0x33	READ_LOG_FILE	Read binary log
0x34	DELETE_LOG_FILE	Delete binary log
0xA4	LOGIN_DEVICE	Re-auth on reconnect
0xB4	SET_PILL_INTERVAL	Set med interval (hours)
0xB5	SET_PILL_INTERVAL_START	Set interval start hour
0xB6	SET_SAFETY_TIMEOUT	Set safety timeout
0xEF	ADD_ADMINKEY	Add admin override key

Responses echo the opcode followed by data, or E:<code>,<message> on error. Success ACKs are [Opcode] ACK.